ARYA College of Engineering & Information Technology

ARYA COLLEGE CLUBSPHERE -A Centralized Web-Based Student Club and Event Management System

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Abstract—This paper presents the design and implementation of Arya College ClubSphere, a centralized web-based application developed to digitize and streamline student club management, activity execution, and event registrations across 26 institutional clubs. Traditionally, extra-curricular campus administration heavily relies on fragmented communication channels such as physical banners, messaging applications, manual registration slips, and individual spreadsheets, leading to information asymmetry, data redundancy, and poor administrative oversight. ClubSphere resolves these operational bottlenecks by providing a role-based architectural environment encompassing four distinct access tiers: Administrator, Club Head, Club Member, and General Student. Built using Java 25, Enterprise Servlets, JavaServer Pages (JSP), Java Database Connectivity (JDBC), and a MySQL relational database deployed on Apache Tomcat 10.1, the system features a dynamic QR-code-based event registration pipeline. The dynamic QR engine enables instant, friction-free registration access for students via smartphones while automating real-time dynamic dashboard updates for administrative analytics. Experimental evaluation demonstrates substantial reductions in manual data processing overhead, complete elimination of physical paperwork, and enhanced student participation visibility across technical, cultural, sports, and social clubs.

Keywords—Java Web Development, JSP & Servlets, Role-Based Access Control (RBAC), QR Code Event Registration, MySQL, Campus Management System, Apache Tomcat.

1. Introduction

Higher educational institutions host a wide array of extra-curricular and co-curricular activities to promote overall student development. At Arya College of Engineering & IT, 26 distinct student clubs span across technical domains, cultural arts, athletic endeavors, social welfare, and creative outlets. Managing such a diverse ecosystem demands efficient communication, precise record maintenance, dynamic scheduling, and seamless participant registration.

Historically, campus activities relied on decentralized and manual methodologies. Event details were disseminated primarily through printed posters, WhatsApp group broadcasts, and verbal announcements. Registrations were collected using manual physical logs or disconnected online forms (e.g., standalone Google Forms). These fragmented approaches introduced significant systemic inefficiencies:

1. Information Asymmetry: Students frequently missed updates regarding upcoming workshops, competitions, or cultural trials due to cluttered messaging channels.

2. Data Redundancy & Corruption: Participant records maintained across separate spreadsheets led to duplicate registrations, inconsistent student identity records, and lost participation entries.

3. Administrative Blind Spots: College faculty coordinators and institutional administrators lacked unified real-time visibility into overall club performance, total active registrations, and inter-club resource utilization.

To solve these systemic challenges, Arya College ClubSphere was conceptualized and developed as a single-point digital platform. The system bridges the gap between campus administration, club management bodies, and the broader student community through role-tailored dashboards and secure database transactions.

1.1 Institutional Overview & Problem Statement

Higher educational institutions serve as vibrant hubs for academic excellence and extra-curricular engagement. At Arya College of Engineering & IT, student development is fostered through a diverse network of 26 dedicated student clubs spanning technical domain societies, robotics groups, cultural performing arts, sports leagues, social service organizations, and competitive coding chapters.

Managing such a vast campus ecosystem involves organizing dozens of workshops, hackathons, guest lectures, cultural trials, and intra-college tournaments throughout the academic calendar. Traditionally, campus club management and event coordination operated under a decentralized, paper-reliant, and fragmented administrative model.

The primary challenges identified in the pre-existing system included:

1. Fragmented Communication Channels: Event announcements were broadcasted across disconnected media including printed posters, WhatsApp group chats, physical notice boards, and social media pages. This produced high information noise, leading to missed registration deadlines and reduced overall student participation.
2. Manual Paperwork & Data Fragmentation: Registration processes relied heavily on manual paper logs or isolated Google Forms created independently by each club coordinator. This fragmented data approach led to data loss, inconsistent student roll number logging, and manual entry errors.
3. Lack of Role-Based Security: Physical attendance sheets and shared spreadsheets lacked operational security, allowing unauthorized edits or missing audit trails for event management teams.
4. Administrative Oversight Inefficiencies: Faculty coordinators and senior college administrators lacked a single unified dashboard to track active participation, monitor budget allocations, assess inter-club resource utilization, or audit event attendance metrics in real-time.

1.2 Proposed Solution: Arya College ClubSphere

To eliminate these administrative bottlenecks, Arya College ClubSphere was designed and built as a centralized, enterprise-grade web platform. The platform unifies all 26 campus clubs under a single web domain, providing role-based multi-tier dashboards for Administrators, Club Heads, Club Members, and Students. Key features include dynamic QR-code-based instant event registrations, automated task delegation workflows, real-time analytics, and localized notification management.

1.3 Research Objectives

The primary objectives of this project and research paper are:

- Architecture Design: To engineer a robust 3-Tier Model-View-Controller (MVC) web application architecture using Java Servlets, JSP, JDBC, and MySQL.
- Security & Access Control: To implement Role-Based Access Control (RBAC) ensuring explicit authorization levels across 4 distinct user tiers.
- Frictionless Registration Flow: To develop a dynamic QR-code payload generator enabling students to complete event enrollments within seconds via mobile devices.
- Administrative Governance: To provide comprehensive analytical dashboards for

institutional leaders to track real-time club health and participation metrics.

2. Proposed System Architecture & User Roles

To overcome the challenges of the manual framework, Arya College ClubSphere is designed as a secure, role-based multi-tier web platform. The system categorizes user interaction into four distinct operational roles, each equipped with explicit authorization rights and customized interfaces:

2.1 Administrator / Professor Tier

The Administrator holds the highest systemic privilege, providing macro-level oversight across all 26 campus clubs. Key administrative capabilities include:

- Comprehensive monitoring of institutional clubs, student memberships, active events, and dynamic registration counts.
- System-wide audit capabilities and authorization of high-priority inter-club events.
- Access to real-time analytical reporting tools for institutional participation assessment.

2.2 Club Head Tier

A Club Head is assigned dedicated management authority over their respective club entity. Responsibilities include:

- Configuring the club profile, uploading official emblems, and publishing structural updates.
- Creating and scheduling upcoming events with parameters such as venue, registration limits, and event categorization (Free/Paid, Individual/Group).
- Utilizing the dynamic QR-code generator engine to publish instant event access links.
- Assigning operational tasks to sub-team members and tracking completion statuses.

2.3 Club Member Tier

Club Members assist in event execution and internal management. Features available to members include:

- Real-time updates on internal task assignments (Pending, In Progress, Completed).
- Access to exclusive member-level announcements and official club schedules.
- Historical logging of past event management contributions.

2.4 Student Tier

The broader student population interacts with the platform as participants. Key features include:

- Exploring the central repository of all 26 active college clubs.
- Scanning event QR codes to access immediate, friction-free registration portals.
- Managing personal registration histories and receiving real-time event alerts.

2.1 Multi-Tier Operational Architecture

To replace the inefficient manual workflow, Arya College ClubSphere is designed as a multi-tier, secure enterprise web application. The platform enforces strict Role-Based Access Control (RBAC) to segment administrative capabilities, event control, and general student participation across four primary operational tiers:

2.1.1 Administrator / Faculty Coordinator Tier

The Administrator tier represents the highest authority in the institutional hierarchy. Designed specifically for college faculty coordinators and institutional directors, this tier provides complete macro-level oversight:

- Institutional Dashboard: Real-time aggregated statistics across all 26 active campus clubs, total registered student profiles, active event schedules, and cumulative registration metrics.
- Global Management: Authority to approve or suspend new club registrations, update faculty advisor assignments, and modify global system configurations.
- Inter-Club Audit Capabilities: Ability to review financial logs, venue allocation schedules, and

participation statistics to prevent schedule collisions between major institutional events.
- Advanced Reporting Engine: Exporting comprehensive analytics and participation summaries in tabular formats for institutional accreditation and annual reporting.

2.1.2 Club Head / Lead Organizer Tier
Each of the 26 student clubs is managed by designated Club Heads who possess isolated operational privileges bounded strictly to their respective club entities:
- Club Profile Customization: Managing club descriptions, leadership team rosters, core values, official badges, and contact channels.
- Event Creation & Lifecycle Controller: Configuring event details including title, venue, start/end dates, max seating capacity, registration fees (if applicable), and event tags (Hackathons, Fests, Cultural, Sports).
- Dynamic QR Generation Engine: Instantly synthesizing scannable QR payload markers linked directly to unique event registration endpoints.
- Task Delegation Framework: Assigning internal preparation duties to Club Members with clear timelines and monitoring execution status.

2.1.3 Club Member Tier
Club Members serve as execution teams responsible for event logistics, registration verification, and ground management:
- Internal Workspace: Accessing dedicated task management boards to track assigned responsibilities (Pending, In Progress, Completed).
- Attendance Verification Portal: Scanning student tickets or inputting registration IDs at event venues to mark live attendance.
- Internal Notices: Accessing member-only announcements, execution guidelines, and team communication schedules.

2.1.4 General Student Tier
The broader student population engages with the platform as participants through a streamlined, self-service portal:
- Centralized Directory Exploration: Browsing all 26 institutional clubs, viewing upcoming event calendars, and reviewing past event archives.
- Frictionless QR Registration: Scanning dynamic QR codes displayed on physical banners or digital posters to jump directly to one-click event enrollment forms.
- Personal Participation Tracker: Viewing active event tickets, past attendance records, and receiving real-time status updates via system notifications.

3. System Design & Functional Modules
To ensure high performance, modular scalability, and robust data isolation, Arya College ClubSphere is partitioned into eight comprehensive functional modules:

3.1 Module 1: Authentication & RBAC Engine
The Authentication Engine handles system access and dynamic routing. User credentials are validated against hashed records stored in the relational database. Upon successful authentication, an HTTP session object is initialized with role-based tokens (ADMIN, CLUB_HEAD, MEMBER, STUDENT). Custom Servlet filters intercept every incoming request to enforce access control lists, ensuring users only access authorized routes.

3.2 Module 2: Centralized Club Directory
This module manages metadata across all 26 campus clubs. It supports relational mapping for club profiles, official emblems, faculty advisor details, active student leadership, and historical archives. Students can search, filter, and discover clubs based on domain categories (Technical, Cultural, Sports, AI & Robotics).

3.3 Module 3: Event Lifecycle Management

Club Heads use this module to manage the full event lifecycle. It includes form input validation, venue conflict checking, date range limits, and dynamic updates (e.g., Postponed, Registration Open, Sold Out). Changes automatically update the central student portal.

3.4 Module 4: Dynamic QR Code Registration Engine
This engine generates dynamic QR code graphics for each event. The backend creates a secure URL with the sanitized Event ID and authorization payload. An embedded Java library processes this string into a Base64 PNG image stream, displayed on physical posters and digital banners. Scanning it opens the pre-filled enrollment portal directly.

3.5 Module 5: Internal Task Delegation & Tracking
This module manages internal club workflows for event execution. Club Heads can create tasks, set deadlines, and assign them to Club Members. Members update task statuses in real time (Pending, In Progress, Completed), providing clear operational visibility.

3.6 Module 6: Notification & Communication Subsystem
This subsystem delivers dynamic, context-aware notifications to users. It handles alerts for registration confirmations, venue changes, upcoming deadlines, and official announcements, keeping students and coordinators updated.

3.7 Module 7: Administrative Analytics & Reporting Engine
The Analytics Engine aggregates database metrics to display visual summaries. Administrators can view total student participation, active event counts, club growth trends, and engagement statistics across departments. Reports can be exported for annual institutional records.

3.8 Module 8: Database & Persistence Layer
The database layer manages secure transactions using Java Database Connectivity (JDBC) with prepared statements. It enforces third normal form (3NF) relational constraints to prevent redundancy and ensure thread safety during peak traffic.

4. Technology Stack & Architectural Framework

4.1 Architectural Design Pattern (MVC)
Arya College ClubSphere uses a classic 3-Tier Model-View-Controller (MVC) design pattern. This ensures complete separation between UI rendering, application logic, and database persistence:
- View (Presentation Layer): Built using JSP (JavaServer Pages), HTML5, CSS3, and JavaScript/Bootstrap 5 for interactive, mobile-responsive user interfaces across desktop and mobile devices.
- Controller (Application Logic): Developed using Java Servlets (Java EE 17) running on Apache Tomcat 10.1 server to process incoming HTTP requests, enforce RBAC permissions, and forward responses.
- Model (Persistence Layer): Powered by MySQL 8.0 database engine, communicating with Java servlets via JDBC (Java Database Connectivity) with connection pooling.

4.2 Technology Stack Overview
- Language & Core Engine: Java (JDK 17)
- Web Application Framework: Java Servlets & JSP
- Application Server: Apache Tomcat 10.1
- Relational Database: MySQL 8.0
- Persistence Driver: JDBC (Java Database Connectivity)
- Design & Styling: Bootstrap 5, Custom CSS, JavaScript (ES6)

4.3 Relational Database Entities (3NF Schema)

To ensure zero data redundancy and maximum transaction safety across 26 institutional clubs, the database is normalized to 3NF with key tables:
1. Users (`users`): Stores user credentials, hashed passwords, institutional email, department, and assigned RBAC role (ADMIN, CLUB_HEAD, MEMBER, STUDENT).
2. Clubs (`clubs`): Stores information for all 26 campus clubs including club name, category, faculty advisor, and assigned Club Head ID.
3. Events (`events`): Contains title, description, venue, scheduled date/time, max seating capacity, sponsoring club ID, and generated QR payload token.
4. Event Registrations (`registrations`): Ledger linking registered students to events with live attendance verification flags.

4.4 System Data Flow & Component Architecture

The architectural framework governs how data moves seamlessly across the 3-tier structure:
1. Presentation Tier (Client Layer): User actions (such as submitting an event registration form or scanning a QR payload) trigger HTTP GET or POST requests sent from browser clients to the Apache Tomcat server.
2. Web Container Tier (Servlet/Controller Layer): Apache Tomcat receives HTTP requests and maps them to specialized Servlet controllers (e.g., `EventController`, `RegistrationController`, `AuthController`). The Servlets validate session credentials, check RBAC privileges, and parse parameters.
3. Data Access Tier (JDBC/DAO Layer): Servlet controllers instantiate Data Access Objects (DAOs) which prepare SQL statements. JDBC connection pools execute these queries against the MySQL relational database, returning result sets that are converted into Java bean objects.
4. View Rendering Layer: The controller forwards Java model attributes to JSP views, which dynamically generate HTML/CSS responses sent back to the client browser.

4.5 Deployment Environment & Server Configuration

- Web Server & Container: Apache Tomcat 10.1 configured with HTTPS SSL/TLS encryption for secure network communication across campus networks.
- JDBC Connection Pool Configuration: Implemented via Apache DBCP to manage database connections efficiently, handling up to 200 concurrent user requests during peak event registration hours without performance degradation.
- Cloud / Local Hosting Parameters: Deployed on a Linux-based cloud virtual server with automated database backup routines executed daily.

5. Implementation Details & Code Architecture

5.1 Role-Based Authentication & Session Filter

Security and operational access control in ClubSphere are enforced using Java EE Servlet Filters. Every incoming request is intercepted to verify HTTP session validity and user authority tokens before granting access to protected servlet endpoints.

5.1.1 Authentication Servlet Filter Logic

When a user attempts to access routes such as `/admin/dashboard` or `/club/manage-events`, the `AuthenticationFilter` evaluates session tokens:

```java
@WebFilter("/*")
public class AuthenticationFilter implements Filter {
    public void doFilter(ServletRequest request, ServletResponse response, FilterChain chain)
        throws IOException, ServletException {
            HttpServletRequest req = (HttpServletRequest) request;
            HttpServletResponse res = (HttpServletResponse) response;
    HttpSession session = req.getSession(false);
```

```
        String requestURI = req.getRequestURI();
          boolean isLoggedIn = (session != null &&
session.getAttribute("userRole") != null);

        if (isLoggedIn) {
                                String role = (String)
session.getAttribute("userRole");
                 if (requestURI.contains("/admin/") &&
!"ADMIN".equalsIgnoreCase(role)) {
                 res.sendRedirect(req.getContextPath() +
"/access-denied.jsp");
                 return;
              }
              chain.doFilter(request, response);
        } else {
                 res.sendRedirect(req.getContextPath() +
"/login.jsp");
        }
     }
}
```

6. System Implementation & Dynamic Workflows

6.1 End-to-End System Workflow Execution

The operational flow of Arya College ClubSphere connects students, club organizers, and faculty administrators through synchronized, role-based interaction loops. Upon logging into the portal, users are dynamically routed to their designated workspaces based on authenticated credentials.

6.2 User Interface & Core Dynamic Workflows

6.2.1 Portal Navigation & Centralized Club Exploration

Students can explore all 26 campus clubs, review upcoming hackathons, fests, and workshops, and check live seating capacity from a unified interface.

6.2.2 Role-Based Authentication Interface

The authentication pipeline enforces Role-Based Access Control (RBAC) across four user tiers (Admin, Club Head, Club Member, Student) to guarantee isolated database views and function security.

6.3 Dynamic Event Registration & QR Generation Workflow

1. Event Creation: A Club Head publishes an event, triggering auto-generation of a unique QR payload token in MySQL.
2. QR Matrix Rendering: The backend converts the token into a scannable Base64 PNG image stream embedded in promotional graphics.
3. Mobile Scanning & Enrollment: Students scan the dynamic QR code with a smartphone camera, jumping directly to the enrollment screen.
4. Database Commit: JDBC executes a transaction in 3NF schema, registering student details and generating a live ticket.

7. Testing, Performance Evaluation & Results

7.1 Software Testing & Authorization Validation

To validate system security, operational stability, and execution speed across all 26 campus clubs, an extensive multi-tier testing process was conducted. Tests included unit testing, RBAC permission verification, dynamic QR flow execution, and concurrent database transaction handling.

7.1.1 Execution Log & Functional Test Cases

Table 1: Key Functional & Security Test Execution Log

Test Case ID	Target Module	Input / Action	Expected Outcome	Status
TC-AUTH-01	RBAC Security	Student attempting Admin Dashboard	Request blocked, 403 redirected	PASS

TC-AUTH-02 | Session Handler | Session timeout after 15 mins inactivity| Force logout to login page | PASS
TC-CLUB-01 | Directory Engine | Search "Coding" in club portal | Filters 26 clubs to Tech group | PASS
TC-EVT-01 | Event Lifecycle | Club Head creating event with past date | Form validation error shown | PASS
TC-QR-01 | QR Engine | Mobile scan of dynamic QR token | Instant redirect to ticket page | PASS
TC-REG-01 | Concurrency Control | Concurrent registration at max capacity | Rollback via JDBC transaction | PASS

7.2 Performance Benchmarks & Stress Load Analysis
Simulated load testing using Apache JMeter evaluated system scalability on a high-throughput campus network during peak registration hours.

Table 2: Latency & Throughput Benchmarks under Simulated Load
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Concurrent Users | Target Endpoint | Avg Response Time (ms) | Success Rate (%) | CPU Utilization
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50 Users | Homepage & Directory | 142 ms | 100.0% | 12.4%
150 Users | Event Browsing & Details | 210 ms | 100.0% | 28.1%
300 Users | Dynamic QR Ticket Scan | 325 ms | 99.8% | 45.6%
500 Users | Concurrent Registration | 480 ms | 99.4% | 68.2%

7.3 Quantitative Impact & Key Findings
- Registration Speedup: Average student registration duration dropped from 4.5 minutes (manual paper entry) to under 8 seconds using dynamic QR scanning.
- Redundancy Elimination: Zero duplicate registration records due to strictly enforced primary and unique key database constraints (`user_id`, `event_id`).
- Real-Time Oversight: Faculty coordinators gained immediate visibility into participation metrics across all 26 campus clubs.

8. Conclusion & Future Work

8.1 Conclusion
The development and integration of Arya College ClubSphere mark a structural transformation in how campus organizations, extra-curricular events, and student engagement records are managed. Prior to this platform, institutional workflows across all 26 campus clubs relied on decentralized communication, physical paper logs, and manual Google Forms entries. This legacy model presented severe administrative blind spots, high vulnerability to record duplication, venue scheduling conflicts, and substantial manual labor overhead for faculty coordinators.

Arya College ClubSphere solves these systemic operational bottlenecks by deploying a robust, highly available 3-Tier Model-View-Controller (MVC) web application architecture. Built strictly using Java EE (Servlets, JSP), Apache Tomcat 10.1, and a 3NF normalized MySQL 8.0 relational database, the system enforces complete data isolation and strict Role-Based Access Control (RBAC) across four distinct operational tiers: Administrator, Club Head, Club Member, and General Student.

Key achievements of the implemented platform include:
1. Frictionless Registration via Dynamic QR Engine: Replacing manual sign-ups with dynamic,

dynamically rendered QR payload tickets reduced average student event registration latency from 4.5 minutes down to less than 8 seconds.
2. Zero Administrative Data Redundancy: Enforcing strict relational foreign key constraints and atomic JDBC database transactions eliminated duplicate registrations and phantom participant records.
3. Complete Institutional Visibility: Centralized administrative dashboards provide real-time participation analytics, total club growth metrics, and cross-club activity monitoring for college directors and faculty advisors.
4. Scalable & Low-Latency Execution: Stress testing via Apache JMeter verified that the application handles up to 500 concurrent student transactions with average response times under 480 ms, demonstrating system readiness for peak campus event drives.

8.2 Future Scope & Technological Enhancements

While the current deployment of Arya College ClubSphere fulfills all foundational requirements for automated club management, future development cycles will introduce advanced technological capabilities to scale the platform further:

8.2.1 Native Cross-Platform Mobile Application

Future iterations will expand beyond web browsers by deploying a native mobile application built using React Native or Flutter. This mobile client will enable push notifications for upcoming deadlines, offline QR validation for event check-ins in low-connectivity campus areas, and biometric user authentication (Fingerprint / Face ID) for streamlined security.

8.2.2 Automated PDF Certificate Generation Engine

An automated certificate synthesis module will be integrated into the backend pipeline. Upon event closure and live attendance verification by Club Members, the server will dynamically overlay student metadata onto pre-designed certificate templates, generating verifiable, digitally signed PDF certificates that are instantly emailed to participants and available for download in their personal portal archives.

8.2.3 AI-Driven Personalized Event Recommendation Pipeline

To further improve student engagement across lesser-known academic and non-academic clubs, machine learning algorithms will analyze historical participation data, student department affiliations, and expressed domain interests. The system will then suggest personalized event recommendations and club joining invitations directly on the student home dashboard.

8.2.4 Automated Budgeting & Resource Management Engine

Future releases will incorporate a centralized financial and logistics ledger. Club Heads will be able to submit budget proposals, venue allocation requests, and equipment requisitions directly to faculty administrators for real-time digital approval, eliminating paper-based approval forms completely.

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